Wavy Walls May Help Homes Stand Up to Hurricane Winds

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Engineers in Florida tested a cool idea for **storm** safety: give walls and roofs gentle waves instead of flat sides.

They 3D-printed small building models. Some were flat. Others had waves of different sizes. In a **wind tunnel**, the wavy models felt less wind force. The deepest waves cut peak **pressure** near corners by about 40 to 60 percent. Corners are where wind often does the most damage.

On a flat roof, wind can make one big **swirl** that sucks upward, almost like peeling a sticker. On a wavy surface, the air skips along and makes many smaller **swirls**, so the strong **suction** is weaker.

Pedro Fernández-Cabán, a civil engineer at Florida State University, said unusual shapes can reduce damage in severe weather. The team still needs more tests. If the idea works on real houses, wavy designs could help communities that face strong **storms**.

Word Treasure

storm -- Very strong weather with powerful winds that can damage buildings.

wind tunnel -- A lab machine that blows air over models to test wind forces.

pressure -- How hard the wind pushes or pulls on a surface.

swirl -- A spinning pocket of air that can suck upward on a roof.

suction -- An upward pull caused by low pressure above a surface.

Background you need

Hurricane winds push hardest on the corners and edges of buildings.

Engineers use wind tunnels to test how air moves over house models.

Changing a building's shape can change how strongly the wind pulls on it.

Break it down

WHO: Engineers at Florida universities
WHAT: Testing wavy building shapes against strong winds
WHERE: Wind-tunnel labs using 3D-printed models
WHY: They hope to reduce hurricane wind damage on homes

Why it matters

Strong storms can hurt homes; smarter shapes could help keep families safer.

Test what you remember

1. What shape did engineers test on buildings?

- (A) Cube cars
- (B) Wavy walls and roofs
- (C) Square balloons only
- (D) Triangle socks

2. Where did they test the models?

- (A) Under a lake only
- (B) Inside a fridge
- (C) In outer space
- (D) In a wind tunnel

3. How much could peak corner pressure fall?

- (A) One million percent
- (B) About 40 to 60 percent
- (C) Exactly 1 percent
- (D) Zero percent

4. What can a big vortex on a flat roof create?

- (A) Strong suction
- (B) Free candy
- (C) Warm snow
- (D) Quiet music

5. Who is a co-author quoted in the story?

- (A) A fashion model
- (B) Pedro Fernández-Cabán
- (C) A soccer coach
- (D) A chef

6. What do researchers still plan?

- (A) To ban all houses
- (B) To only paint walls pink
- (C) More tests and computer models
- (D) To stop science

Answer key (try the quiz first!): 1: B 2: D 3: B 4: A 5: B 6: C

Questions to think about

1. Pedro Fernández-Cabán: Says unusual shapes can reduce damage and help designers.

MY THOUGHTS (at least 20 words):